

Measurement Ready

Use the tool, scale, unit, and calculation as one evidence chain.

Name: _____ Date: _____ Class Period: _____

LEARNING TARGETS

Match common lab tools to the quantity and unit they measure. • Choose an appropriate tool and justify the choice with visual evidence. • Calculate density and percent error with correct units and reasonable precision.

Three tools, three different jobs



Electronic balance

Measures mass
Typical unit: g



Graduated cylinder

Measures liquid volume
Typical unit: mL



Beaker

Holds and mixes
Volume is approximate

Measurement rules that prevent avoidable errors

- 1 Record the value and the unit together.
- 2 For an analog scale, record all certain digits plus one estimated digit.
- 3 Read a concave liquid meniscus at eye level from its lowest point.
- 4 Keep units through every calculation and round only the final result.

FORMULA TOOLBOX

density = mass / volume

percent error =
|experimental - accepted| / accepted × 100

Units tell the story

mass: g
liquid volume: mL
density: g/mL

Choose the Tool

Name what you see, then use visible features to justify the choice.

Name: _____ Date: _____ Class Period: _____

A. Identify each instrument

1



Instrument: _____

Quantity/job: _____

Unit (if measured): _____

2



Instrument: _____

Quantity/job: _____

Unit (if measured): _____

3



Instrument: _____

Quantity/job: _____

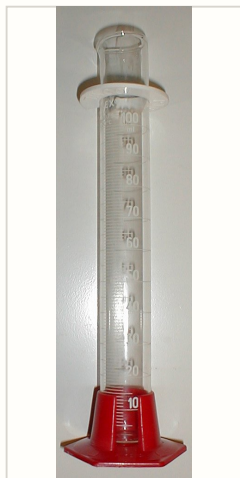
Unit (if measured): _____

B. Select and justify

4. You need 18.6 mL of water. Choose a beaker or a graduated cylinder. Explain which visible design features support your choice.

5. You need about 125 mL of solution and room to stir. Choose a beaker or a graduated cylinder. Explain.

C. Read visual evidence



6. Identify two visible features that show this vessel was designed to measure liquid volume.

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Calculate with Units

Show the formula, substitute values with units, and round only the final answer.

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GUIDED EXAMPLE

A sample has a mass of 36.4 g and a volume of 14.0 mL.

$$d = m / V = 36.4 \text{ g} / 14.0 \text{ mL} = 2.60 \text{ g/mL}$$

7. A liquid sample has a mass of 27.0 g and a density of 2.70 g/mL. Calculate its volume.

Work space

8. A liquid has a density of 0.789 g/mL and a volume of 25.0 mL. Calculate its mass.

Work space

9. A student measures 2.58 g/mL. The accepted value is 2.70 g/mL. Calculate percent error.

Work space

10. Accuracy and precision

Accepted density: 2.70 g/mL

Set A	2.70, 2.71, 2.69 g/mL
Set B	2.42, 2.41, 2.42 g/mL

Which set is accurate? Which is precise? Explain using the pattern in the data.



Reason from Evidence

Use the data to build a claim, diagnose the error, and plan a repair.

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LAB CASE: UNKNOWN LIQUID

A student uses an electronic balance and a graduated cylinder. The empty container has a mass of 42.31 g. The container plus liquid has a mass of 67.19 g. The liquid volume is 20.0 mL. The accepted density is 1.25 g/mL.

11. Mass of liquid: _____

12. Experimental density: _____

13. Percent error: _____

14. Claim - Evidence - Reasoning

Claim: Is the experimental result close to the accepted value?

Evidence: Cite at least two numbers from the case.

Reasoning: Explain what percent error tells you and what it does not tell you about precision.

15. Error diagnosis

Another student reads the liquid above eye level, rounds every intermediate value, and omits units. Name two problems and one exact repair action.

REPAIR LOG

Item/skill to repair: _____

Next action: _____

Incorrect idea or step: _____

When I will retry: _____